ALAHMEDS STUDY / MACROECONOMICS

Midterms 1-7, 10-11 | Formulas and graph rules

One-page landscape study reference. Coverage: 9 of 28 modules. Generated 11 Nov 2026.

Prices and real output [M1]

$\pi = (P_{new}/P_{old} - 1) \times 100\%$

P = price index; π = inflation for the stated period.

Nominal GDP = $\Sigma\, p_t q_t$

Real GDP = $\Sigma\, p_{base} q_t$

p price; q quantity; Σ sum. Real GDP holds base prices fixed.

Fixed-basket index = $(cost_t/cost_{base}) \times 100$

Hold basket quantities fixed. Index 106 → price ratio 1.06.

GDP deflator = (nominal GDP/real GDP) × 100

Divide nominal GDP by deflator/100 to obtain base-year dollars.

$g_{real} = (1+g_{nominal})/(1+\pi_{GDP}) - 1$

Decimals; approx.: real ≈ nominal growth − inflation.

SRC-011 slides 52-57; SRC-079 pp.1-5; SRC-129 p.2.

Buying power and interest [M1]

Goods bought = dollars / (dollars per good)

Fixed dollars: a higher price buys fewer goods.

$r^e \approx i - \pi^e$; $r^a \approx i - \pi^a$

i = nominal rate; e expected; a actual. Match periods.

$1+r^a = (1+i)/(1+\pi^a)$

Exact; decimal rates. Surprise inflation benefits fixed-debt borrowers, hurts lenders.

SRC-011 slides 59-62; SRC-129 p.2. Exact ratio: derived.

GDP and capital investment [M2,5]

Y = C + I + G + NX; NX = EX − IM

C consumption; I physical investment; G purchases; EX exports; IM imports.

I: new fixed capital + homes + Δinventories

Δ change. Transfers are not G; asset trades are not new output.

K′ = (1−d)K + I

K′ next capital; d depreciation share; I new real investment. Match periods.

Value added = output − intermediate inputs

GDP = value added = final output = spending = income. Count once.

Imported phone: C +500, NX −500 ⇒ ΔY = 0

Domestic services count separately. Identity ≠ causal prediction.

S = Y − C − G = I + NX; NX = S − I

S = national saving. All quantities are flows per period.

SRC-011 slides 34-42,47-50; SRC-081 pp.1-3; SRC-140 Q1; SRC-012 slide 13.

Employment denominators [M3]

LF = E + U; pop = E + U + outside

E employed; U jobless, searching. pop = eligible civilian non-institutionalized people age 16+.

Unemployment = U/LF × 100%

Denominator LF. Search exit lowers U and LF; lower rate need not mean more jobs.

Participation = LF/pop × 100%

Denominator pop (people), not price level P. LF includes E and U.

Employment/population = E/pop × 100%

Household: people; payroll: covered jobs. Zero cyclical can leave frictional/structural unemployment.

SRC-011 slides 25-31. pop is teaching shorthand.

Marginal products and hiring [M4,6]

MPN = $0.7\, A(K/N)^{0.3} = 0.7Y/N$

Marginal output/hour. Local gain; whole-hour gain is approximate.

MPK = $0.3\, A(N/K)^{0.7} = 0.3Y/K$

Own input ↑ → own MP ↓. K ↑ → MPN ↑; N ↑ → MPK ↑ (other inputs fixed).

Competitive firm: MPN = w = W/P

MPN > w: hire more; MPN < w: reduce hours. Fixed A,K: w ↑ → N demanded ↓.

Labor income = 0.7Y; capital income = 0.3Y

Competitive marginal-product payments divide the same Y. A or K ↑ shifts labor demand right.

SRC-012 slides 4-7,18-20; SRC-082; SRC-129 pp.3-7.

Labor supply: movements and shifts [M7,10]

w ↑ → N_s ↑, PVLR fixed; PVLR ↑ → N_s ↓, w fixed

PVLR = present value of lifetime resources. Wage movement versus wealth shift; normal leisure.

Temporary tax ↑: N_s ←; N* ↓; w* ↑; R ↓

Set lifetime-resource effect aside; A,K fixed; no offsetting transfer.

Permanent tax ↑: substitution ←; income →

Normal leisure; no offsetting transfer. Net N* can ↓, stay or ↑. R lower at exact offset.

Population ↓ → aggregate N_s ←

Remaining individual supply schedules can be unchanged.

Value of leisure ↑ → individual + aggregate supply ←

Demand fixed: N* ↓, w* ↑. Wage alone moves along supply; fixed A,K holds demand fixed.

SRC-012 slides 21-27,32-35; SRC-083 pp.2-8,11; SRC-131.

Production and growth [M4-5]

$Y = A \times K^{0.3} \times N^{0.7}$

Y output/period; K capital stock; N hours/period; A efficiency.

$ALP = Y/N = A(K/N)^{0.3}$

ALP = output/hour. Fixed A,K: N ↑ → Y ↑, ALP ↓.

$Y'/Y = (A'/A)(K'/K)^{0.3}(N'/N)^{0.7}$

Exact growth factors; prime = new. Fixed A: scale K,N together → same scale for Y.

$g_Y \approx g_A + 0.3g_K + 0.7g_N$

g = growth rate. Small-change approximation; match periods and rate units. Use exact factors for large changes.

$g_A \approx g_Y - 0.3g_K - 0.7g_N$

TFP growth is the residual; A ≠ output/hour. Higher K/N raises ALP at fixed A.

$g_{Y/N} \approx g_A + 0.3(g_K - g_N) \approx g_Y - g_N$

Approximate output/hour growth. g = growth rate; match periods and rate units.

SRC-012 slides 4-8,12-15; SRC-082; SRC-129 pp.3-7. Ratios derived.

Work reward and total hours [M7,10]

w = W/P; R = $[(1-t_n)/(1+t_c)]w$

W nominal dollars/hour; P dollars/good. w gross, R net goods/hour. t_n labor tax; t_c sales tax; decimals.

w=10: labor tax 25% → 7.5; sales tax 25% → 8

Separate cases: 10×.75 vs 10/1.25. P excludes sales tax.

$N_s(w) = \Sigma_j h_j(w)$; equal h: N_s = pop×h

At the same w, sum individual hours. pop people; h hours/person.

$N_d(w^*) = N_s(w^*) = N^*$

* = equilibrium; N = total hours, not people.

SRC-012 slides 21-27,32-35; SRC-083 pp.2-8,11. Tax ratio/sum derived.

Potential output and graph axes [M11]

$Y^* = A\, K^{0.3}(N^*)^{0.7}$

N* clearing hours; Y* potential output, not a physical maximum.

Labor: W/P vertical; N horizontal

Demand slopes down; course supply up, lifetime resources fixed.

LRAS: P vertical; Y horizontal; Y = Y*

P price level. Flexible W: double P and W → same W/P,N*,Y*.

A,K or clearing N* changes → LRAS can shift

P alone: along LRAS. Sticky W can put Y below Y*. Lower clearing N* alone ≠ cyclical unemployment.

SRC-012 slides 4,8,19-29,52-59; SRC-129 p.13.

Exam: one handwritten standard-size sheet, front only; no photocopy. This PDF is a study/hand-copy guide. [SRC-127]